

TrustLens

Product Requirements Document (PRD) *A Real-Time Trust Scoring Engine for Digital Platforms*

Team iostream

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1. Product Overview

TrustLens is an AI-powered trust scoring engine that analyses any digital platform across five independent signals and produces a single, explainable Trust Score (0–100). Think of it as a credit score for data privacy — except it works for platforms, not people.

The Problem

Users interact with hundreds of digital platforms daily with zero visibility into how responsibly those platforms handle their personal data. Existing tools are either too technical, outdated, or rely on self-reported compliance claims. India's DPDP Act 2023 and GDPR have created regulatory accountability structures, but users still lack a real-time, accessible trust signal at the point of interaction.

The Solution

TrustLens gives users a single, independently generated, plain-English score for any platform — powered by five automated signals: privacy policy analysis, breach history, regulatory compliance, user complaint patterns, and third-party tracker detection.

2. Goals & Success Metrics

Goal	Success Metric
Deliver a working demo in hackathon timeframe	Pre-loaded scores for 5 platforms live on demo day

Show real contrasting results	Min 30-point gap between best and worst demo platform
Be India-aware	DPDP compliance badge visible on all results
Impress judges with UX	Score + breakdown visible in under 3 seconds
Make it actionable	Every result includes personalised user action steps

3. Complete Feature List

3.1 Trust Score Engine (Core)

- Composite score 0–100 across 5 weighted signals
- Grade system: A (80–100) / B (60–79) / C (40–59) / D (20–39) / F (0–19)
- Colour-coded grade badge: Green / Blue / Yellow / Orange / Red
- "Last Updated" timestamp on every score card
- Score trend indicator — ☐ improved / ☐ declining (based on historical delta)
- Scores cached in PostgreSQL for instant repeat lookups

3.2 Signal 1 — Privacy Policy Analysis (20 pts)

- Auto-fetch and scan the platform's privacy policy via URL scraping
- Red-flag clause detection (e.g. "we may share with partners", "we retain data indefinitely")
- Positive signal detection (e.g. explicit data minimisation, clear deletion policy)
- Plain-English summary of the 3 biggest risks found
- Powered by Gemini API (free tier, 1500 req/day)
- Sub-score displayed with reason for each deduction

3.3 Signal 2 — Data Breach History (30 pts)

- Check if platform domain has been in any recorded breach (HIBP free /breaches endpoint)
- Show breach date, number of users affected, and data types exposed
- Severity scoring: HIGH / MEDIUM / LOW based on recency and scale
- Visual breach detail card:
 - Platform name, breach date, user count affected
 - Data exposed: Emails, Passwords, Phone numbers, etc.
 - Severity label with point penalty shown
- Historical breach timeline if multiple breaches exist

- No API key required — uses HIBP public /breaches endpoint

3.4 Signal 3 — Regulatory Compliance (20 pts)

- Structured compliance checklist against India's DPDP Act 2023
- GDPR alignment check for platforms serving EU users
- Checklist items evaluated:
 - Clear consent mechanism present
 - Data deletion / erasure request process exists
 - Data Protection Officer (DPO) mentioned
 - Data retention period clearly stated
 - Third-party sharing disclosed with opt-out
- Prominent DPDP Compliance Badge: ✓Compliant / ✗Non-Compliant
- Unique India-specific differentiator — no other tool does this

3.5 Signal 4 — User Complaint Analysis (15 pts)

- Aggregate privacy-related complaints from app store reviews and consumer forums
- HuggingFace Inference API for sentiment + topic classification
- Classify complaints by category: data misuse, spam, unauthorised sharing, tracking
- Show complaint volume and top complaint themes
- High complaint volume on privacy topics = score penalty

3.6 Signal 5 — Third-Party Tracker Detection (15 pts)

- Detect trackers using Exodus Privacy open dataset (downloadable CSV)
- Cross-reference Mozilla Privacy Not Included listings
- Show exact tracker names and their parent companies
- Plain-English output: "Instagram has 12 trackers sending your data to Facebook, Google, and 3 data brokers"
- More trackers + data broker partnerships = higher penalty

3.7 Score Breakdown Dashboard

- Colour-coded per-signal breakdown with sub-score and max score
- Progress bar for each signal (visual score representation)
- Expandable detail cards per signal (click to see full reasoning)
- Overall grade badge prominently displayed at top

- DPDP Badge displayed alongside grade

3.8 Trend Graph

- Line chart showing trust score over last 12 months
- Visual  /  direction arrow with delta value
- Narrative label: "This platform's trust is declining" or "Improving"
- Pre-loaded historical data for demo platforms

3.9 "What You Should Do" Section

- Actionable recommendations generated based on that platform's specific weaknesses
- 3–5 plain-English steps per platform
- Examples:
 - Turn off ad personalisation in settings
 - Revoke third-party app permissions
 - Enable two-factor authentication
 - Download your data and review what's stored
- Makes TrustLens feel like an assistant, not just a scorecard

3.10 Compare Mode

- Search two platforms side by side
- Winner highlighted with  badge
- Per-signal comparison with colour-coded winner per category
- Plain-English summary: "Swiggy wins on tracker detection and breach history"
- Extremely demo-friendly and visually impressive for judges

3.11 Search & Results

- Search any platform by name from the dashboard
- Cached scores return instantly (PostgreSQL lookup)
- New platforms trigger live analysis pipeline
- Search suggestions for pre-loaded platforms

3.12 Pre-Loaded Demo Platforms

- 5 platforms pre-scored with full breakdown for demo day:
 - Zomato — food delivery, Indian platform
 - Instagram — social media, global
 - BYJU'S — EdTech, Indian platform
 - Aarogya Setu — health/government app
 - Swiggy — food delivery (Compare Mode demo partner)
- Contrasting scores to show full grade range to judges
- Historical trend data pre-loaded for each platform

3.13 Chrome Extension (WOW Factor — Prototype/Mockup)

- Browser toolbar badge showing TrustScore when visiting a website
- Green (A/B) / Yellow (C) / Red (D/F) colour indicator
- Click badge to see quick summary popup
- Even a working prototype or high-fidelity mockup will impress judges massively
- Shows real-world application of TrustLens beyond the dashboard

4. Scoring System

4.1 Signal Weights

Signal	Weight / Max Points
Privacy Policy Analysis	20 pts
Data Breach History	30 pts (increased — most objective signal)
Regulatory Compliance (DPDP + GDPR)	20 pts
User Complaint Patterns	15 pts
Third-Party Tracker Detection	15 pts
TOTAL	100 pts

4.2 Grade System

Score Range	Grade & Meaning
80 – 100	A — Trustworthy ✓ Strong privacy practices
60 – 79	B — Acceptable □ Minor concerns

40 – 59	C — Caution ☐ Notable risks present
20 – 39	D — Risky ☐ Multiple red flags
0 – 19	F — Avoid ☒ Serious violations

4.3 Sample Score Card Output

Instagram — Trust Score: 52/100 Grade: C ☐

Privacy Policy  14/20 Vague data sharing language
 Breach History  24/30 No major breaches recorded
 DPDP Compliance  8/20 Non-Compliant **✗**
 User Complaints  7/15 High volume of privacy complaints
 Tracker Detection  9/15 12 trackers detected

DPDP Badge: ✗Non-Compliant

Trend: ☐ Score dropped 9 pts in 6 months

What You Should Do:

- Disable ad personalisation in Instagram settings
- Revoke third-party app permissions
- Enable two-factor authentication

5. Tech Stack

Layer	Technology	Cost	Notes
Frontend	React + Tailwind CSS	Free	Component-based UI
Backend	FastAPI (Python)	Free	Async REST API
Policy Analysis	Gemini API	Free	1500 req/day free tier
Breach Data	HIBP /breaches endpoint	Free	No API key needed
Tracker Detection	Exodus Privacy Dataset	Free	Download CSV once
Sentiment Analysis	HuggingFace Inference API	Free	Free token, rate-limited
Scoring Engine	scikit-learn	Free	Weighted composite model

Database	PostgreSQL	Free	Cache scores
Deployment	Railway	Free	Free hobby tier

6. System Architecture

6.1 Request Flow

1. User types platform name in React frontend
2. React sends GET /score?platform=zomato to FastAPI backend
3. FastAPI checks PostgreSQL — is there a cached score?
YES → return instantly
NO → trigger all 5 signal modules in parallel
4. Each module returns a sub-score (0–30 depending on weight)
5. scikit-learn scoring engine combines them into final score
6. Result stored in PostgreSQL for future cache hits
7. JSON response returned to React
8. React renders: score, grade, breakdown, trend graph, action steps

6.2 Backend Module Breakdown

Module	Responsibility
policy_analyzer.py	Fetch privacy policy URL, send to Gemini API, parse red flags, return sub-score
breach_checker.py	Call HIBP /breaches, filter by domain, score severity + recency
compliance_checker.py	Run DPDP/GDPR checklist, return sub-score and badge status
complaint_analyzer.py	Call HuggingFace sentiment API on scraped reviews, return complaint score
tracker_detector.py	Load Exodus CSV, match platform, count trackers, return sub-score
scoring_engine.py	Combine all 5 sub-scores into weighted composite, assign grade
cache_manager.py	PostgreSQL read/write for score caching

7. Hackathon Build Plan

Day 1 — Backend Core

- Set up FastAPI project structure and folder layout
- Implement `scoring_engine.py` with weighted model
- Implement `breach_checker.py` using HIBP free endpoint
- Implement `tracker_detector.py` using Exodus CSV
- Implement `compliance_checker.py` with DPDP checklist
- Set up PostgreSQL and `cache_manager.py`
- Test all 5 modules independently with dummy data

Day 2 — AI Signals + Frontend

- Integrate Gemini API for `policy_analyzer.py`
- Integrate HuggingFace Inference API for `complaint_analyzer.py`
- Build React dashboard — search bar, score card, breakdown
- Connect frontend to FastAPI backend
- Add trend graph component
- Add Compare Mode UI

Day 3 — Polish + Demo Prep

- Pre-load all 5 demo platforms with full data
- Add "What You Should Do" action section
- Add DPDP badge and score trend indicators
- Deploy to Railway
- Build Chrome extension mockup (or working prototype)
- Rehearse demo flow for judges
- Prepare answers for judge Q&A

8. Demo Script for Judges

Follow this exact sequence on demo day for maximum impact:

1. Open TrustLens dashboard. Show clean search interface.
2. Search "Aarogya Setu" — show a high score (A grade). Explain DPDP compliance badge.
3. Search "BYJU'S" — show a low score (D grade). Walk through breach card + complaint themes.
4. Use Compare Mode: Zomato vs Swiggy. Show per-signal winner highlighting.
5. Show the trend graph for Instagram — declining 9 pts over 6 months.
6. Show Chrome extension badge on a live website (or mockup).

Anticipated Judge Questions

Question	Answer
Can platforms game the score?	Policy analysis is only 20%. The other 80% comes from independent third-party data — breach databases, tracker datasets, user complaints — which platforms cannot control.
Is the data real-time?	Breach and tracker data updates when new sources are released. Policy analysis is triggered fresh on every new search. Cached scores show “Last Updated” timestamp.
Why is DPDP compliance important?	India's DPDP Act 2023 is newly enacted. TrustLens is one of the only tools that scores platforms specifically against this law, making it uniquely relevant for the Indian market.
How is this different from existing tools?	Most privacy tools tell you WHAT data is collected. TrustLens tells you HOW responsibly the platform behaves with it — across 5 independent automated signals, not self-reporting.
Can this scale beyond a hackathon?	Yes. The scoring engine is modular — new signals can be added. The Chrome extension creates a B2C distribution channel. At scale, platforms face market pressure to improve scores.

9. Accounts & API Keys Needed

Sign up for these before starting to build. All are free.

Service	URL	What You Get
Google AI Studio	aistudio.google.com	Gemini API key — 1500 free req/day
HuggingFace	huggingface.co	Free Inference API token
Railway	railway.app	Free deployment hosting
HIBP	haveibeenpwned.com/api/v3/breaches	No signup needed — public endpoint
Exodus Privacy	reports.exodus-privacy.eu.org	Download tracker dataset CSV

10. Open Questions / Decisions

Decision	Options
Chrome extension scope	Full working extension vs high-fidelity Figma mockup for demo day

Complaint data source	Scrape live app store reviews vs use pre-collected static dataset for hackathon
Trend graph data	Pre-loaded fake historical data vs calculate from real breach dates
Policy fetch method	Auto-scrape from URL vs paste policy text manually for demo platforms
Hosting database	Railway PostgreSQL (free) vs SQLite for simplicity in hackathon

TrustLens — One Line Mission

TrustLens should feel like a doctor giving a diagnosis — not just a number, but what's wrong,
how serious it is, and exactly what to do about it.

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